SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

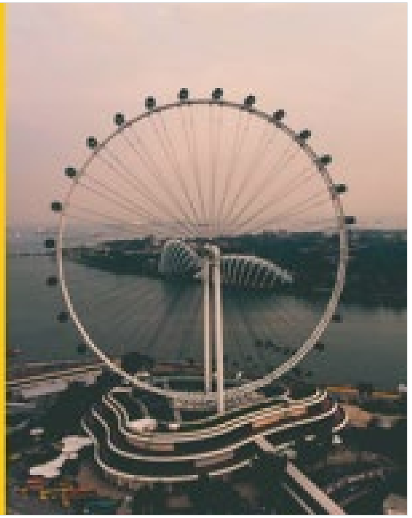
Beginner

TRIGONOMETRIC FUNCTIONS

UNIT CIRCLE APPROACH

Session outline

I. The Unit Circle



II. Trigonometric functions of real numbers

Refer to the following pages in the textbook!

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 5,
Sections 5.1-5.4

Learning Objectives

- Understand the **Unit Circle** and how it is used to define trigonometric functions.
- Review and understand the concept of **coterminal angles** and how to determine them.
- Determine Signs of Trigonometric Functions
- Practice locating points on the unit circle based on given coordinates.
- Learn how to evaluate trigonometric functions using the unit circle and reference angles.
- Review the properties **of even and odd functions and how they apply to trigonometric functions**.
- Understand and apply fundamental trigonometric identities.
- Write One Trigonometric Function in Terms of Another

I. The Unit Circle

Understanding the two approaches in Trigonometry

Trigonometric functions

Right triangle approach

Trig functions defined using **ratios of the sides** of a right angle triangle

Unit circle approach

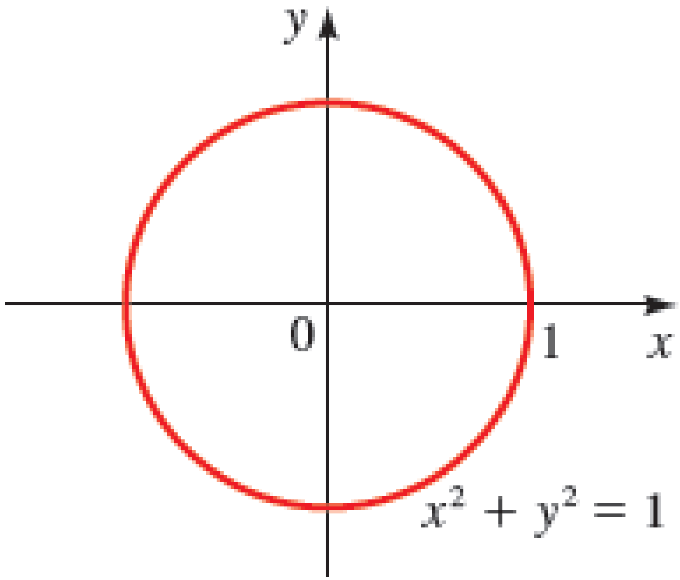
Trig Functions defined using **coordinates of points on the unit circle**

The unit circle

THE UNIT CIRCLE

The **unit circle** is the circle of radius 1 centered at the origin in the *xy*-plane. Its equation is

$$x^2 + y^2 = 1$$



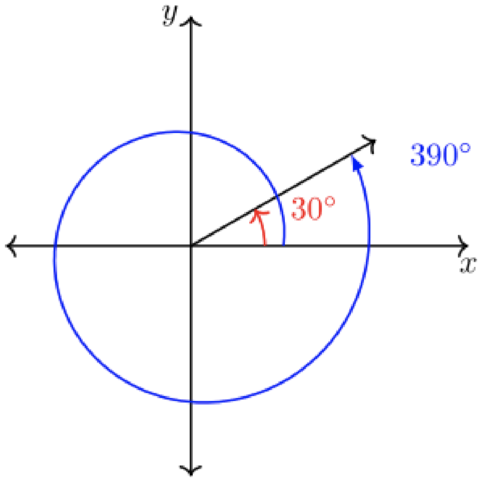
Equation of a circle centred at (0,0) with radius r

$x^2 + y^2 = r^2$

Equation of a circle centred at (0,0) with radius 1

$x^2 + y^2 = 1$

Coterminal angles

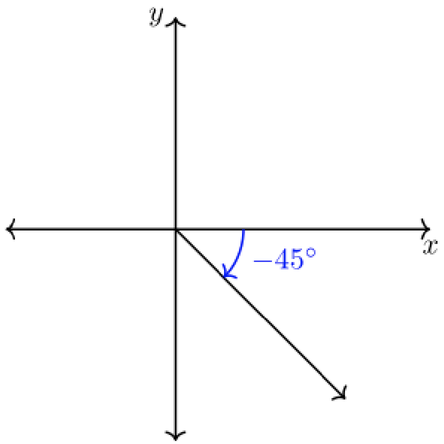


Two or more standard angles that share **common terminal sides** are said to be **coterminal angles**.

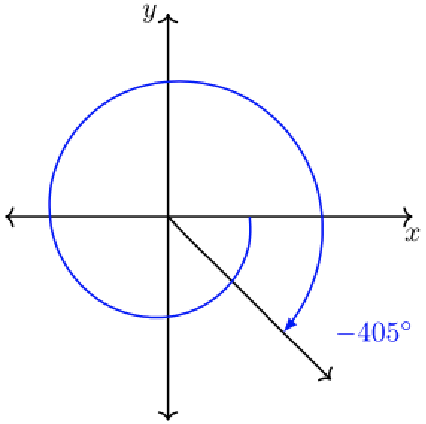
For example, 30 ° and 390 ° are coterminal angles.

Since coterminal angles correspond to **the same point** on the unit circle,

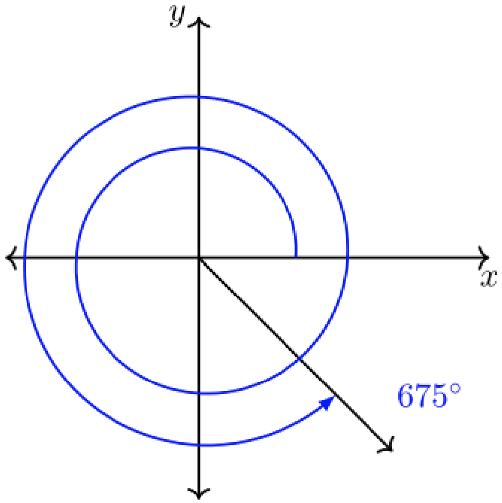
they have the **same values** for trig functions



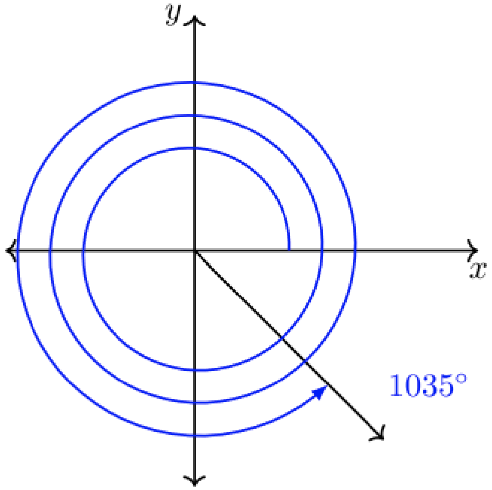
$-45° - 360°$



$360° + 315°$



$2(360°) + 315°$



Each angle shares the same terminal side.
All four angles are coterminal with 315 ° , and coterminal with each other.

REVIEW

Signs of the Trigonometric Functions

Q I: Values of the trigonometric functions are all **positive** if the angle θ has its terminal side in Quadrant I. This is because *x* and *y* are positive in this quadrant.

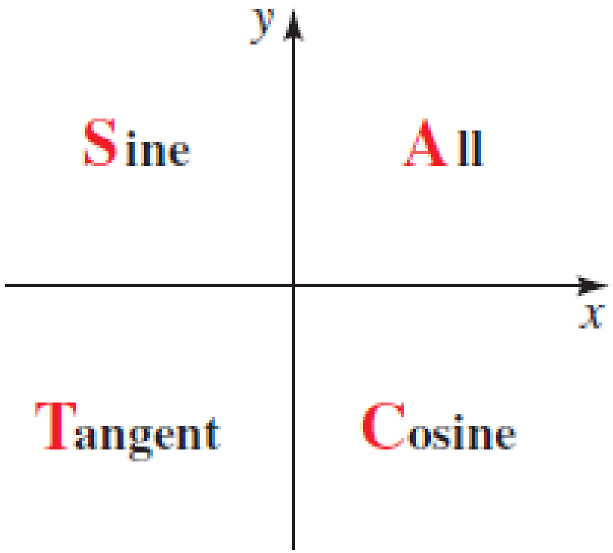
SIGNS OF THE TRIGONOMETRIC FUNCTIONS

Quadrant	Positive Functions	Negative Functions
I	all	none
II	sin, csc	cos, sec, tan, cot
III	tan, cot	sin, csc, cos, sec
IV	cos, sec	sin, csc, tan, cot

QII: Since the ***y* and *r* are both positive**, sine is **positive**.

Since the ***x* is negative** and *r* is positive, cosine is **negative**

Trig functions that are positive in each quadrant



You can remember this as "All Students Take Calculus."

REVIEW

Determining points on the unit circle

Show that the point $P\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3}\right)$ is on the unit circle.

$x^2 + y^2 = \mathbf{1}$ Equation of unit circle

$$\left(\frac{\sqrt{3}}{3}\right)^2 + \left(\frac{\sqrt{6}}{3}\right)^2$$

$$= \frac{3}{9} + \frac{6}{9}$$

$$= 1$$

P is on the unit circle.

Verify that the point **satisfies the equation** of the unit circle!

1. Identify the coordinates
2. Plug in coordinate in the equation
3. Verify if **LHS = RHS**

EXAMPLE

Locating a point on the Unit circle

The point $P(\sqrt{3}/2, y)$ is on the unit circle in Quadrant IV. Find its y-coordinate.

SOLUTION Since the point is on the unit circle, we have

$$\left(\frac{\sqrt{3}}{2}\right)^2 + y^2 = 1$$

$$y^2 = 1 - \frac{3}{4} = \frac{1}{4}$$

$$y = \pm\frac{1}{2}$$

Since the point is in Quadrant IV, its y-coordinate must be negative, so $y = -\frac{1}{2}$.

EXAMPLE

1. Locating points on the unit circle

The point P is on the unit circle. Find $P(x, y)$ from the given information.

(a) The x-coordinate of P is $\frac{5}{13}$, and the y-coordinate is negative

(b) The y-coordinate of P is $-\frac{3}{5}$, and the x-coordinate is positive.

(c) The x-coordinate of P is positive, and the y-coordinate of P is $-\frac{\sqrt{5}}{5}$

(d) The x-coordinate of P is $-\frac{2}{5}$, and P lies above the x-axis.

ACTIVITY 1

1a. Locating points on the unit circle

(a) The x-coordinate of P is $\frac{5}{13}$, and the y-coordinate is negative

We have the following equation for the unit circle:

$x^2 + y^2 = 1$, so we plug in $P(\frac{5}{13}, y)$,

$(\frac{5}{13})^2 + y^2 = 1$

$y^2 = 1 - \frac{25}{169} = \frac{144}{169}$

$y = \frac{12}{13}$ or $y = -\frac{12}{13}$ but the y-coordinate is negative so $y = -\frac{12}{13}$

SOLUTION

1b. Locating points on the unit circle

(b) The y-coordinate of P is $-\frac{3}{5}$, and the x-coordinate is positive.

We have the following equation for the unit circle:
$x^2 + y^2 = 1$, so we plug in $P(x, \frac{-3}{5})$,
$x^2 + (\frac{-3}{5})^2 = 1$
$y^2 = 1 - \frac{9}{25} = \frac{16}{25}$
$y = \frac{4}{5}$ or $y = -\frac{4}{5}$ but the x-coordinate is positive so $x = \frac{4}{5}$

SOLUTION

1c. Locating points on the unit circle

(c) The x-coordinate of P is positive, and the y-coordinate of P is $-\dfrac{\sqrt{5}}{5}$

We have the following equation for the unit circle:

$x^2 + y^2 = 1$, so we plug in $P(x, \frac{\sqrt{5}}{5})$,

$x^2 + (\frac{\sqrt{5}}{5})^2 = 1$

$x^2 = 1 - \frac{5}{25} = \frac{20}{25}$

$x = \frac{2\sqrt{5}}{5}$ or $x = -\frac{2\sqrt{5}}{5}$ but the x-coordinate is positive so $x = \frac{2\sqrt{5}}{5}$

SOLUTION

1d. Locating points on the unit circle

(d) The x-coordinate of P is $-\frac{2}{5}$, and P lies above the x-axis

The x-coordinate of P is $-\frac{2}{5}$, and P lies above the x-axis

Substitute x by $-\frac{2}{5}$ into the equation of the unit circle:

$$x^2 + y^2 = 1$$

$$\left(-\frac{2}{5}\right)^2 + y^2 = 1$$

Solve for y:

$$\frac{4}{25} + y^2 = 1$$

$$y^2 = 1 - \frac{4}{25}$$

$$y^2 = \frac{21}{25}$$

$$\sqrt{y^2} = \sqrt{\frac{21}{25}}$$

$$y = \pm\frac{\sqrt{21}}{5}$$

Since P lies above the x-axis, y must be positive. The point P is:

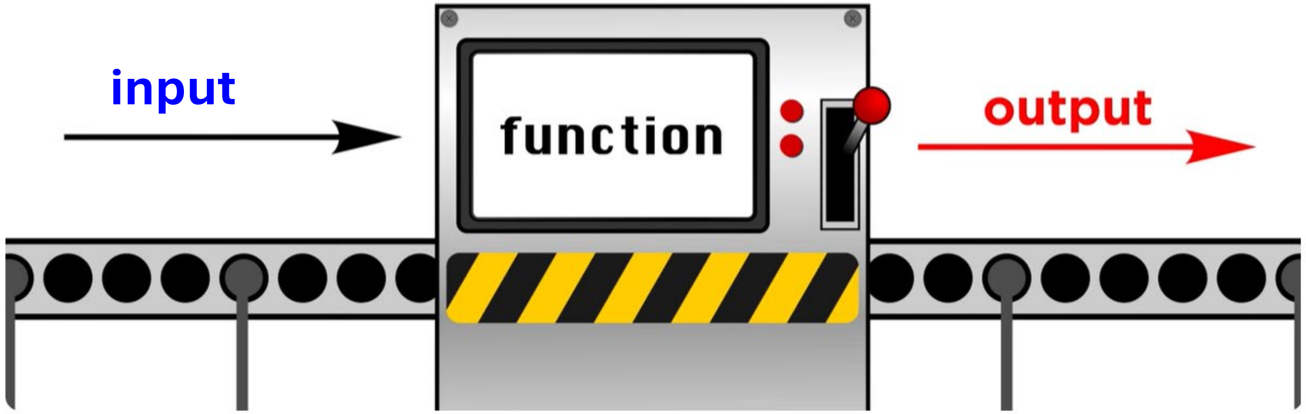
$$P\left(-\frac{2}{5}, \frac{\sqrt{21}}{5}\right)$$

SOLUTION

II. Trigonometric functions of real numbers

What is a function?

A ***function* is a** rule that describes how one quantity depends on another

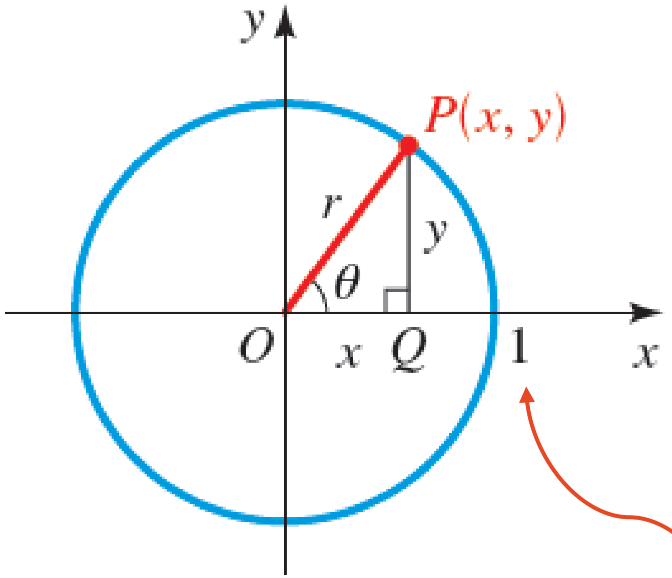


REVIEW

We can also think of a function **like a machine:**

when we insert **input** values,
it spits out the **output** values

Trigonometric functions of real numbers



Triangle OPQ is a right triangle.

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{y}{1} = y$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{x}{1} = x$$

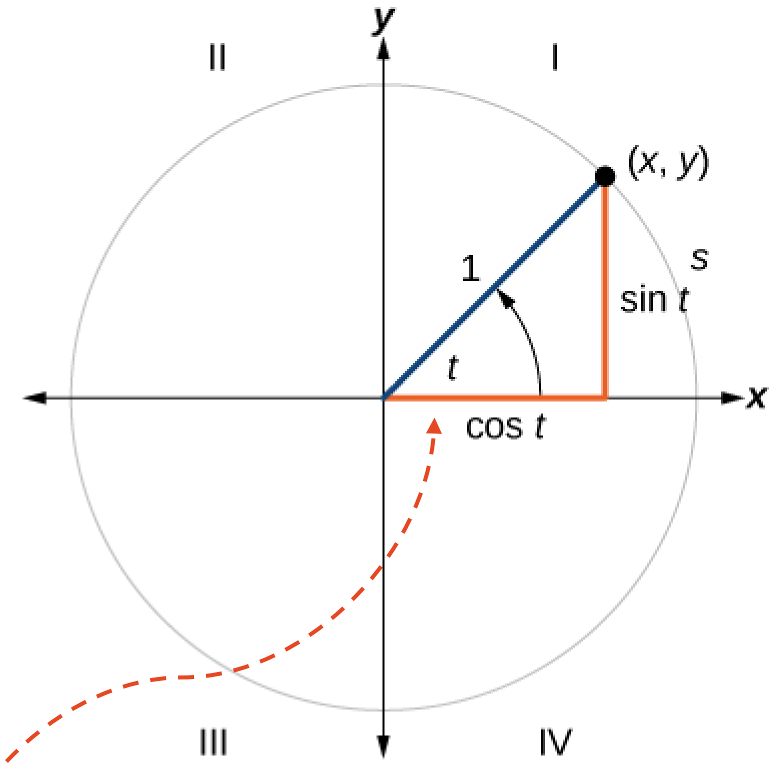
Radius = Hypotenuse= **1**

Unit circle: A circle with radius **1**

REVIEW

(x, y) coordinates as outputs of Trig functions

The unit circle definition allows us to extend the domain of sine and cosine to **all real numbers**.



Unit circle where the central angle is t radians

For any angle t, we can label the intersection of the terminal side and the unit circle as by its coordinates, (x,y).

The coordinates x and y will be the outputs of the trigonometric functions **f(t)=cost** and **f(t)=sint** respectively.

This means

$$x = \cos t$$
$$y = \sin t$$

The coordinates of point (x, y) are (**cost**, **sint**)

REVIEW

Trigonometric **functions**

DEFINITION OF THE TRIGONOMETRIC FUNCTIONS

Let t be any real number and let $P(x, y)$ be the terminal point on the unit circle determined by t. We define

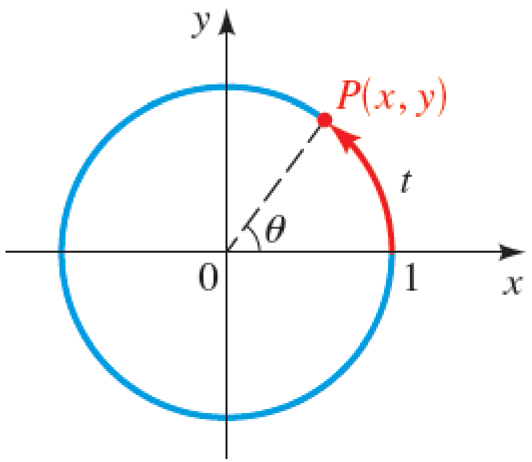
$$\sin t = y \qquad \cos t = x \qquad \tan t = \frac{y}{x} \quad (x \neq 0)$$

$$\csc t = \frac{1}{y} \quad (y \neq 0) \qquad \sec t = \frac{1}{x} \quad (x \neq 0) \qquad \cot t = \frac{x}{y} \quad (y \neq 0)$$

Now, we are thinking of

Sine
Cosine
tangent,
cosecant,
secant,
Cotangent

As **functions** with input **t**

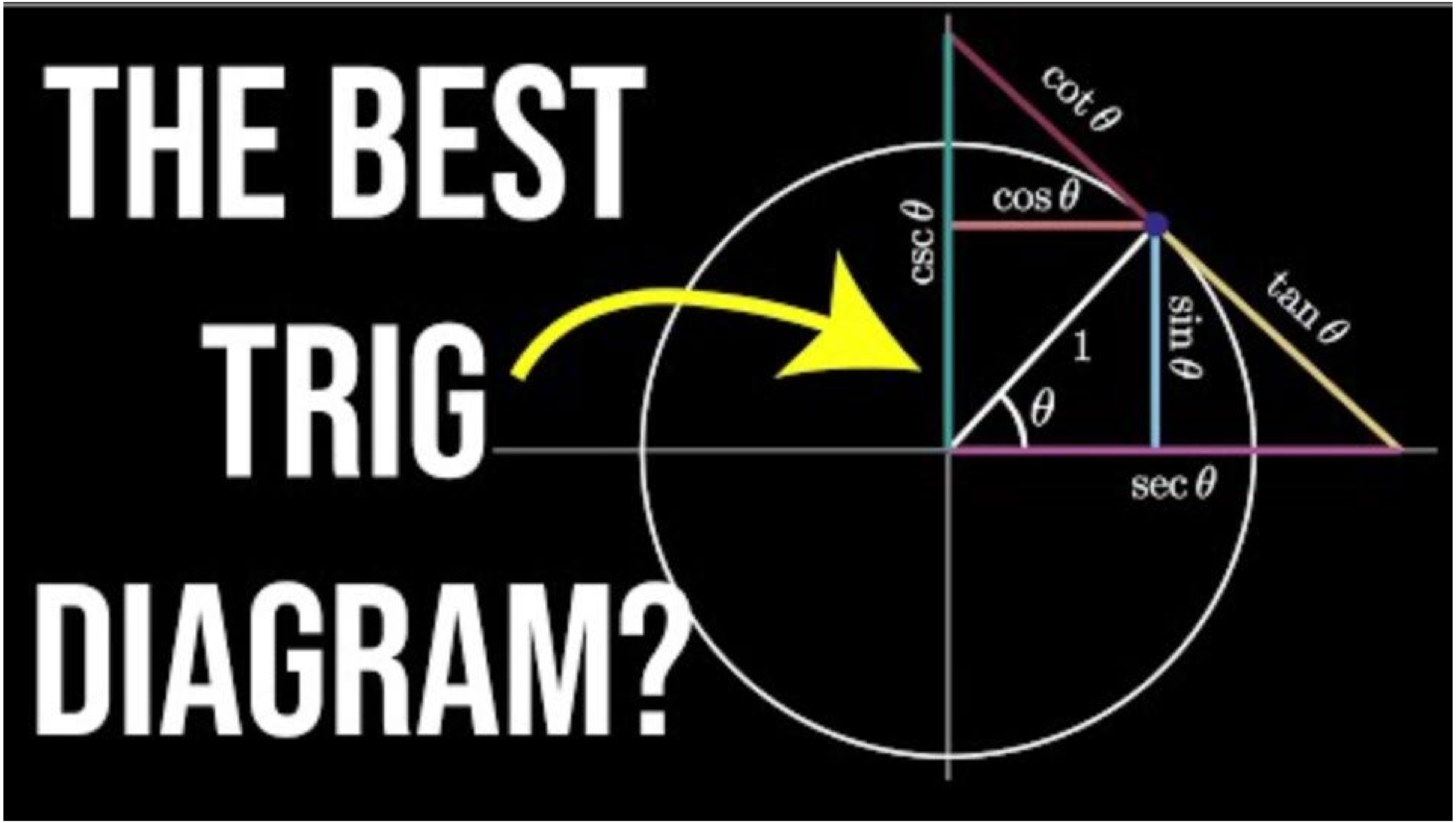


If θ is measured in radians, then $\theta = t$.

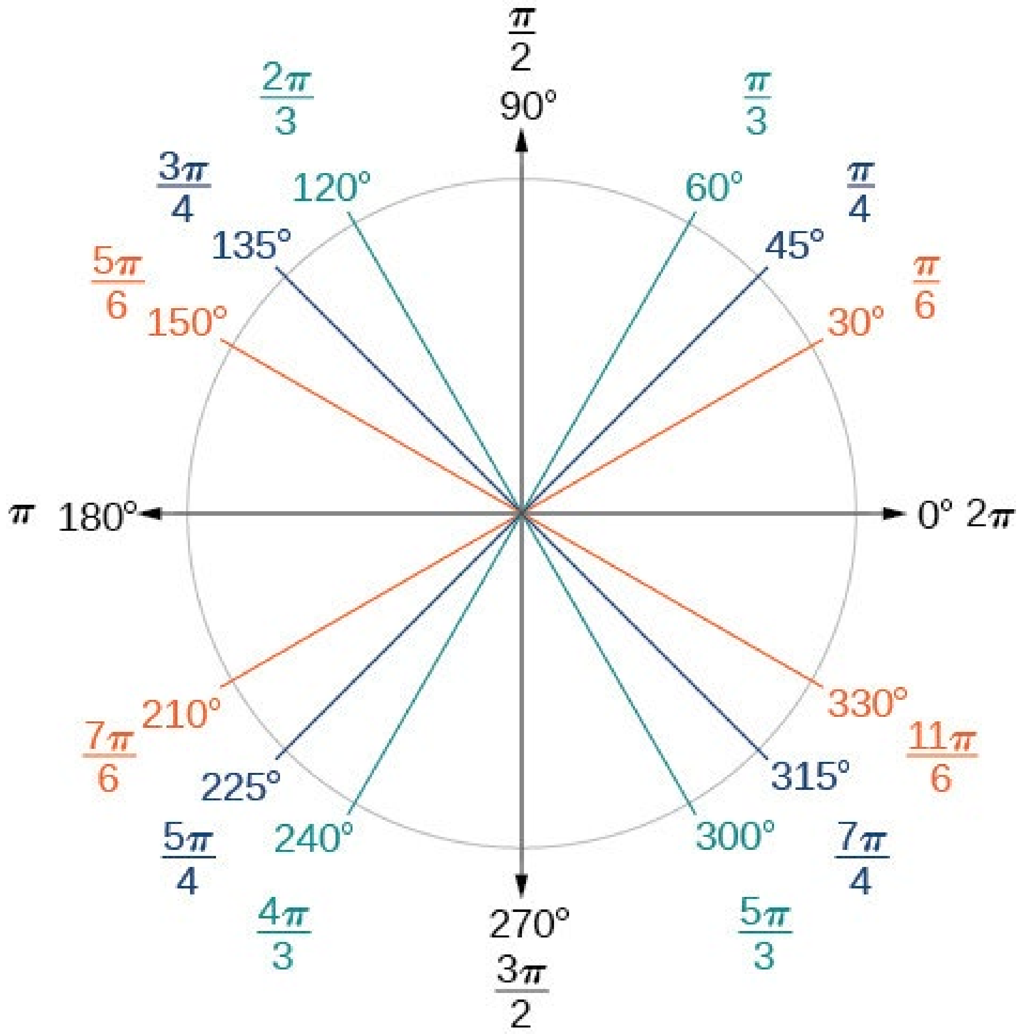
[Because t=rθ, and r=1]

Trigonometric functions are **periodic**, meaning the same values are reproduced with each **360° revolution**.

Trigonometric functions in the unit circle



Commonly used angles in the unit circle



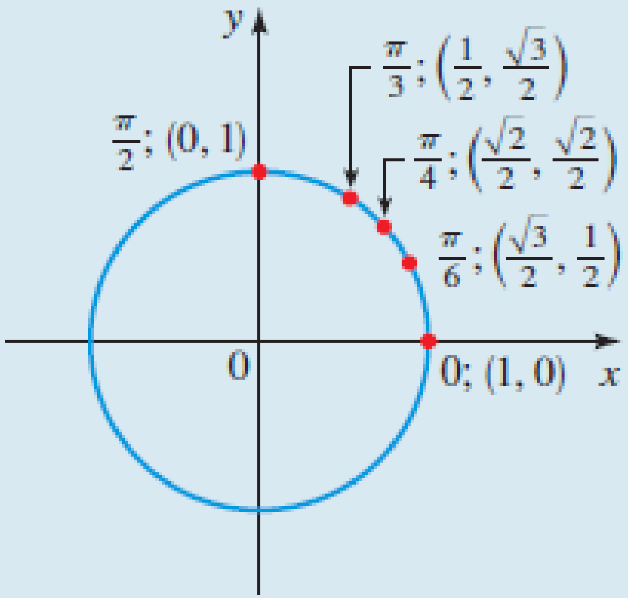
Explore the coordinates and angles of the unit circle here!

https://www.geogebra.org/m/nv9vex3X

Special values of the trigonometric functions

TABLE 1

t	$\sin t$	$\cos t$	$\tan t$	$\csc t$	$\sec t$	$\cot t$
0	0	1	0	—	1	—
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{2}$	1	0	—	1	—	0



Good to memorize these!

We can easily remember the sines and cosines of the basic angles by writing them in the form $\sqrt{\square}/2$:

t	$\sin t$	$\cos t$
0	$\sqrt{0}/2$	$\sqrt{4}/2$
$\pi/6$	$\sqrt{1}/2$	$\sqrt{3}/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$\sqrt{3}/2$	$\sqrt{1}/2$
$\pi/2$	$\sqrt{4}/2$	$\sqrt{0}/2$

Evaluate Trigonometric functions

Evaluate the six trigonometric functions for the given angle:

$$\theta = \pi$$

SOLUTION

Use the angle on the unit circle to find the corresponding x and y-coordinates.
For π, $x = -1$ and $y = 0$

$$\sin \pi = y = 0$$

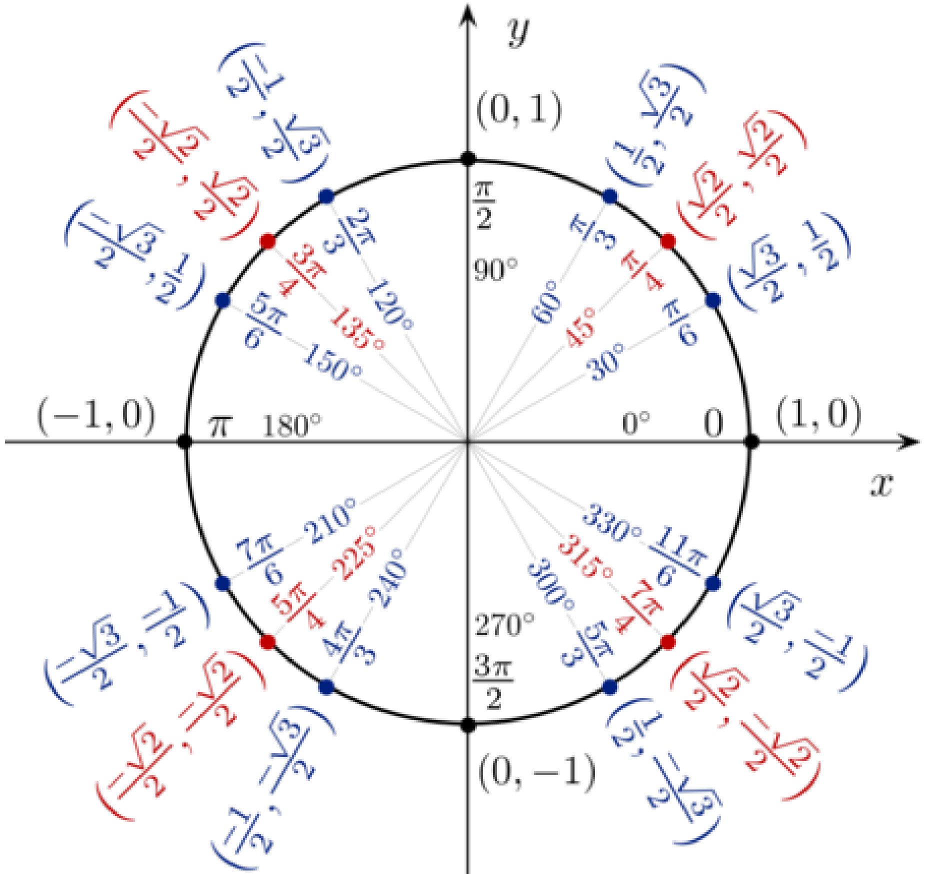
$$\cos \pi = x = -1$$

$$\tan \pi = \frac{y}{x} = \frac{0}{-1} = 0$$

$$\csc \pi = \frac{1}{y} = \frac{1}{0} = \text{undefined}$$

$$\sec \pi = \frac{1}{x} = \frac{1}{-1} = -1$$

$$\cot \pi = \frac{x}{y} = \frac{-1}{0} = \text{undefined}$$



EXAMPLE

Using Reference Angles to Find Trigonometric Functions

1. **Find the reference angle.** Find the reference angle by measuring the smallest angle to the x -axis.

2. **Evaluate function** at the reference angle

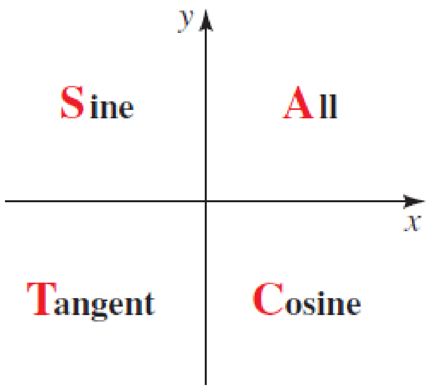
3. **Observe the quadrant** where the terminal side of the original angle is located. Based on the quadrant, determine whether the output is positive or negative.

Using Reference Angles to Find Trigonometric Functions

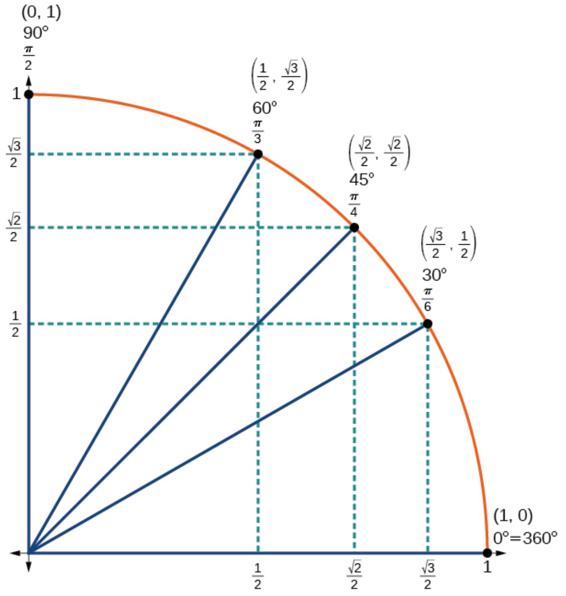
Use reference angles to find all six trigonometric functions of $-\dfrac{5\pi}{6}$

SOLUTION

The angle between this angle's terminal side and the *x*-axis is $\dfrac{\pi}{6}$, so that is the reference angle.

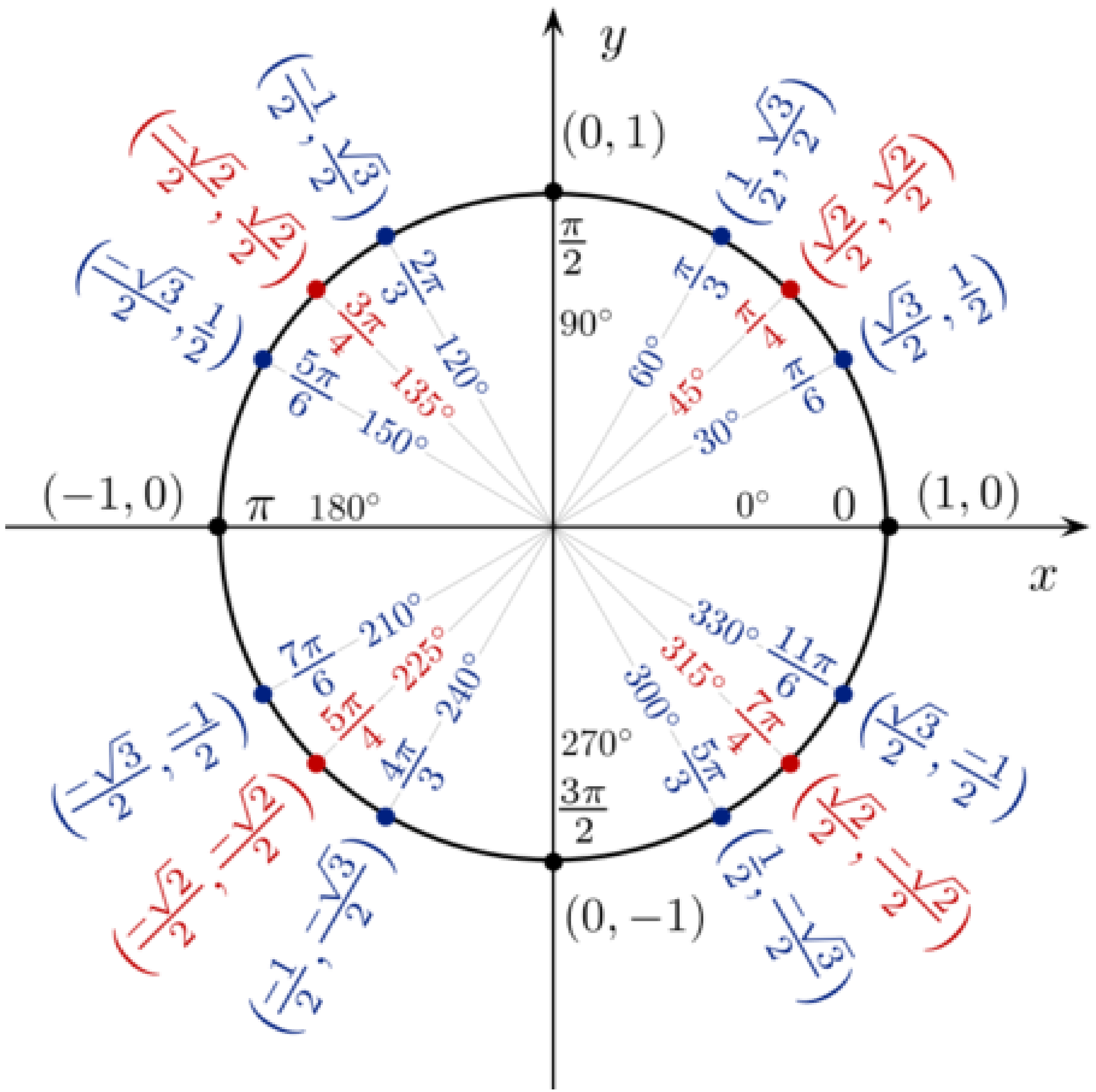


Since $-\dfrac{5\pi}{6}$ is in the third quadrant, where both x and y are negative, cosine, sine, secant, and cosecant will be negative, while tangent and cotangent will be positive.



$$\cos\left(-\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}, \sin\left(-\frac{5\pi}{6}\right) = -\frac{1}{2}, \tan\left(-\frac{5\pi}{6}\right) = \frac{\sqrt{3}}{3}$$
$$\sec\left(-\frac{5\pi}{6}\right) = -\frac{2\sqrt{3}}{3}, \csc\left(-\frac{5\pi}{6}\right) = -2, \cot\left(-\frac{5\pi}{6}\right) = \sqrt{3}$$

EXAMPLE



y
x
(0, 1)
(1, 0)
(−1, 0)
(0, −1)
0
0°
π/2
90°
π
180°
3π/2
270°
π/6
30°
π/4
45°
π/3
60°
2π/3
120°
3π/4
135°
5π/6
150°
7π/6
210°
5π/4
225°
4π/3
240°
5π/3
300°
7π/4
315°
11π/6
330°
(√3/2, 1/2)
(√2/2, √2/2)
(1/2, √3/2)
(−1/2, √3/2)
(−√2/2, √2/2)
(−√3/2, 1/2)
(−√3/2, −1/2)
(−√2/2, −√2/2)
(−1/2, −√3/2)
(1/2, −√3/2)
(√2/2, −√2/2)
(√3/2, −1/2)

2. Evaluate Trigonometric functions

Evaluate the six trigonometric functions for the given angles:

(a) $\theta = \frac{4\pi}{3}$

(b) $\theta = \frac{11\pi}{6}$

(c) $\theta = -\frac{\pi}{3}$

(d) $\theta = \frac{9\pi}{4}$

ACTIVITY 2

2a Evaluate the six Trigonometric functions

(a) $\theta = \dfrac{4\pi}{3}$

SOLUTION

Use the angle on the unit circle to find the corresponding x and y-coordinates. For $\frac{4\pi}{3}$, $x = -\frac{1}{2}$, and $y = -\frac{\sqrt{3}}{2}$

$$\sin \frac{4\pi}{3} = y = -\frac{\sqrt{3}}{2}$$

$$\csc \frac{4\pi}{3} = \frac{1}{y} = \frac{1}{-\frac{\sqrt{3}}{2}} = -\frac{2\sqrt{3}}{3}$$

$$\cos \frac{4\pi}{3} = x = -\frac{1}{2}$$

$$\sec \frac{4\pi}{3} = \frac{1}{x} = \frac{1}{-\frac{1}{2}} = -2$$

$$\tan \frac{4\pi}{3} = \frac{y}{x} = \frac{-\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = \sqrt{3}$$

$$\cot \frac{4\pi}{3} = \frac{x}{y} = \frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{\sqrt{3}}{3}$$

SOLUTION

2b Evaluate the six Trigonometric functions

(b) $\theta = \frac{11\pi}{6}$

Use the angle on the unit circle to find the corresponding *x* and *y*-coordinates.

For $\frac{11\pi}{6}$, $x = \frac{\sqrt{3}}{2}$ and $y = \frac{-1}{2}$

$$\sin\frac{11\pi}{6} = y = -\frac{1}{2}$$

$$\cos\frac{11\pi}{6} = x = \frac{\sqrt{3}}{2}$$

$$\tan\frac{11\pi}{6} = \frac{y}{x} = \frac{-\frac{1}{2}}{\frac{\sqrt{3}}{2}} = -\frac{\sqrt{3}}{3}$$

$$\csc\frac{11\pi}{6} = \frac{1}{y} = \frac{1}{-\frac{1}{2}} = -2$$

$$\sec\frac{11\pi}{6} = \frac{1}{x} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2\sqrt{3}}{3}$$

$$\cot\frac{11\pi}{6} = \frac{x}{y} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$$

SOLUTION

2c Evaluate the six Trigonometric functions

(c) $\boldsymbol{\theta = -\frac{\pi}{3}}$

The first step is to find a coterminal angle between 0 and 2π

$$\theta = -\frac{\pi}{3} + 2\pi$$

$$\theta = -\frac{\pi}{3} + \frac{6\pi}{3}$$

$$\theta = \frac{5\pi}{3}$$

The coordinates for $\theta = \frac{5\pi}{3}$ are $\left(\frac{1}{2}, \frac{-\sqrt{3}}{2}\right)$. Use those in the trigonometric functions and evaluate.

$$\sin\left(-\frac{\pi}{3}\right) = y = -\frac{\sqrt{3}}{2} \qquad \csc\left(-\frac{\pi}{3}\right) = \frac{1}{y} = \frac{1}{-\frac{\sqrt{3}}{2}} = -\frac{2\sqrt{3}}{3}$$

$$\cos\left(-\frac{\pi}{3}\right) = x = \frac{1}{2} \qquad \sec\left(-\frac{\pi}{3}\right) = \frac{1}{x} = \frac{1}{\frac{1}{2}} = 2$$

$$\tan\left(-\frac{\pi}{3}\right) = \frac{y}{x} = \frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = -\sqrt{3} \qquad \cot\left(-\frac{\pi}{3}\right) = \frac{x}{y} = \frac{\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = -\frac{\sqrt{3}}{3}$$

SOLUTION

2d Evaluate the six Trigonometric functions

(d) $\boldsymbol{\theta = \frac{9\pi}{4}}$

The first step is to find a coterminal angle between 0 and 2π.

$$\theta = \frac{9\pi}{4} - 2\pi$$

$$\theta = \frac{9\pi}{4} - \frac{8\pi}{4}$$

$$\theta = \frac{\pi}{4}$$

The coordinates for $\theta = \frac{\pi}{4}$ are $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ Use those in the trigonometric functions and evaluate.

$$\sin\frac{9\pi}{4} = y = \frac{\sqrt{2}}{2} \qquad \csc\frac{9\pi}{4} = \frac{1}{y} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$$

$$\cos\frac{9\pi}{4} = x = \frac{\sqrt{2}}{2} \qquad \sec\frac{9\pi}{4} = \frac{1}{x} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$$

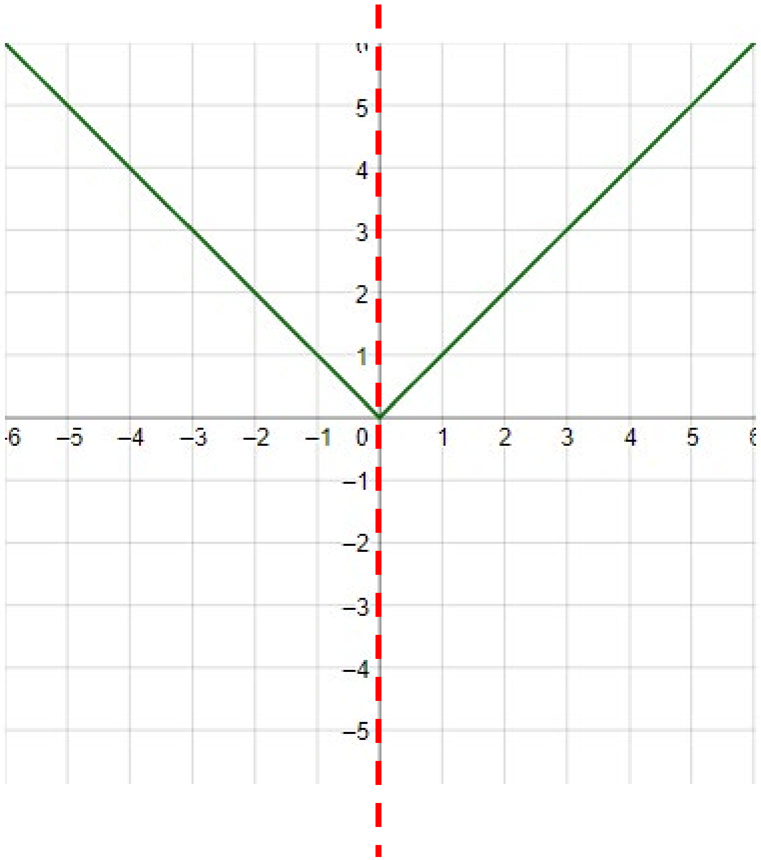
$$\tan\frac{9\pi}{4} = \frac{y}{x} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1 \qquad \cot\frac{9\pi}{4} = \frac{x}{y} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1$$

SOLUTION

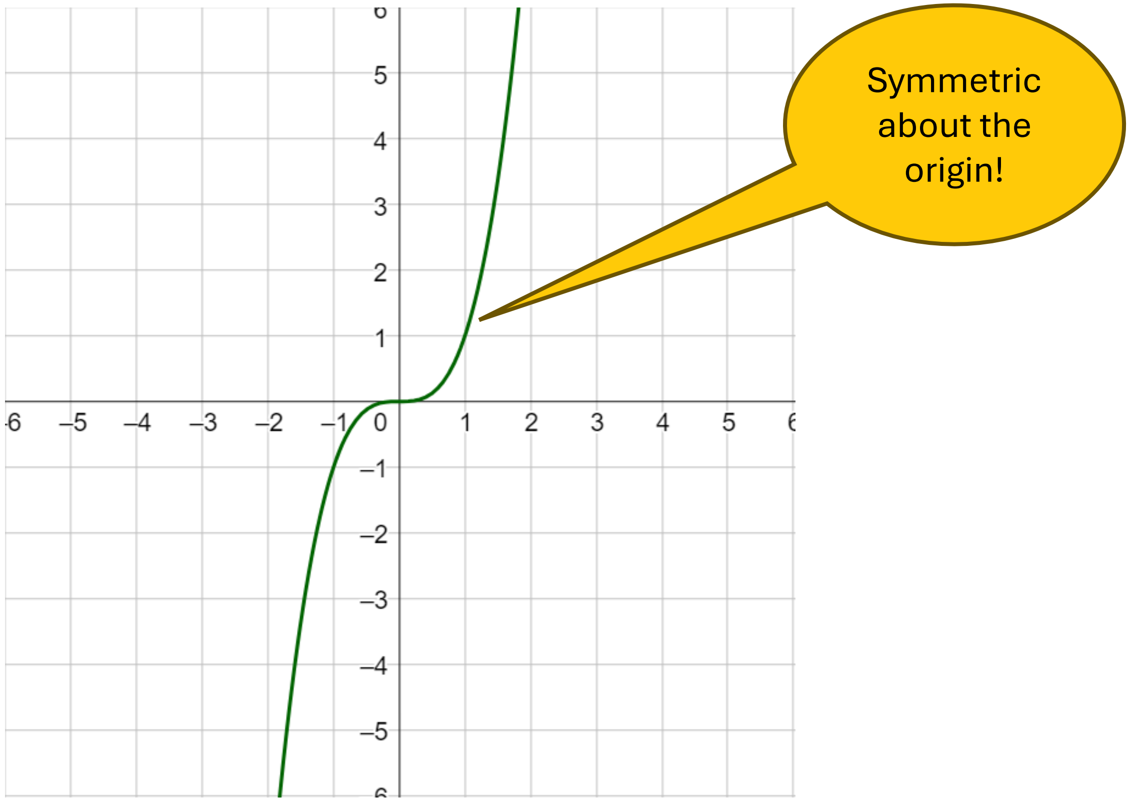
Even and Odd functions

Graphs that are symmetric when reflecting over the *y*-axis are called **even functions**.

Functions that are symmetric after reflecting in both axes, or rotating 180°, are called **odd functions**.



An **even function** is one in which $f(-x)=f(x)$.

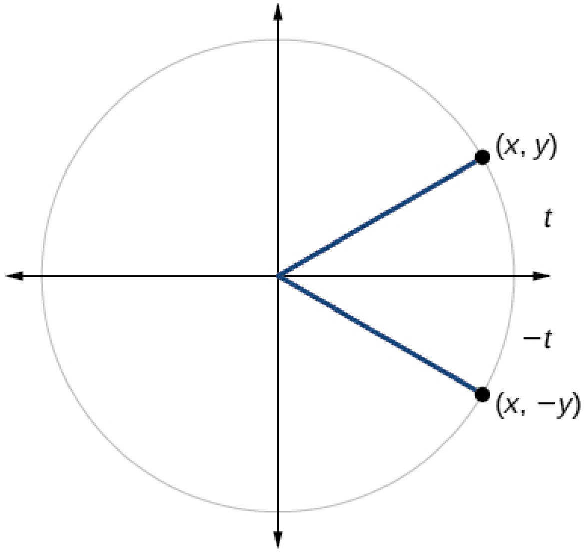


An **odd function** is one in which $f(-x)=-f(x)$.

REVIEW

How do the trig functions treat negative input?

We can test whether a trigonometric function is even or odd by drawing a **unit circle** with a positive and a negative angle



$x = \cos t$
$y = \sin t$

The sine of the **positive** angle is **y** .

The sine of the **negative** angle is **−y**.

$$\sin(-t) = -y = -\sin t$$

Similarly, for the cosine function,

$$\cos(-t) = x = \cos t$$

$$\tan(-t) = \frac{-y}{x} = -\frac{y}{x} = -\tan t$$

Sine and **tangent** are **odd functions**, whereas **cosine** is an **even** function.

Reciprocal of an even function is even and the reciprocal of an odd function is odd.

Even and Odd Properties of Trig functions

EVEN-ODD PROPERTIES

Sine, cosecant, tangent, and cotangent are odd functions; cosine and secant are even functions.

$$\sin(-t) = -\sin t \qquad \cos(-t) = \cos t \qquad \tan(-t) = -\tan t$$

$$\csc(-t) = -\csc t \qquad \sec(-t) = \sec t \qquad \cot(-t) = -\cot t$$

Fundamental Identities

FUNDAMENTAL IDENTITIES

Reciprocal Identities

$$\csc t = \frac{1}{\sin t} \qquad \sec t = \frac{1}{\cos t} \qquad \cot t = \frac{1}{\tan t} \qquad \tan t = \frac{\sin t}{\cos t} \qquad \cot t = \frac{\cos t}{\sin t}$$

Pythagorean Identities

$$\sin^2 t + \cos^2 t = 1 \qquad \tan^2 t + 1 = \sec^2 t \qquad 1 + \cot^2 t = \csc^2 t$$

REVIEW

Even and Odd Trigonometric functions

Use the even-odd properties of the trigonometric functions to determine each value.

(a) $\sin\left(-\frac{\pi}{6}\right)$ (b) $\cos\left(-\frac{\pi}{4}\right)$

SOLUTION

(a) $\sin\left(-\frac{\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}$ Sine is odd

(b) $\cos\left(-\frac{\pi}{4}\right) = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ Cosine is even

EXAMPLE

Finding all Trigonometric functions from the value of one

If $\cos t = \frac{3}{5}$ and t is in Quadrant IV, find the values of all the trigonometric functions at t.

SOLUTION From the Pythagorean identities we have

$$\begin{aligned} \sin^2 t + \cos^2 t &= 1 \\ \sin^2 t + \left(\tfrac{3}{5}\right)^2 &= 1 && \text{Substitute } \cos t = \tfrac{3}{5} \\ \sin^2 t &= 1 - \tfrac{9}{25} = \tfrac{16}{25} && \text{Solve for } \sin^2 t \\ \sin t &= \pm\tfrac{4}{5} && \text{Take square roots} \end{aligned}$$

Since this point is in Quadrant IV, $\sin t$ is negative, so $\sin t = -\frac{4}{5}$. Now that we know both $\sin t$ and $\cos t$, we can find the values of the other trigonometric functions using the reciprocal identities.

$$\sin t = -\frac{4}{5} \qquad \cos t = \frac{3}{5} \qquad \tan t = \frac{\sin t}{\cos t} = \frac{-\frac{4}{5}}{\frac{3}{5}} = -\frac{4}{3}$$

$$\csc t = \frac{1}{\sin t} = -\frac{5}{4} \qquad \sec t = \frac{1}{\cos t} = \frac{5}{3} \qquad \cot t = \frac{1}{\tan t} = -\frac{3}{4}$$

EXAMPLE

Writing One Trigonometric Function in Terms of Another

EXAMPLE

Write $\tan t$ in terms of $\cos t$, where t is in Quadrant III.

SOLUTION Since $\tan t = \sin t/\cos t$, we need to write $\sin t$ in terms of $\cos t$. By the Pythagorean identities we have

$$\sin^2 t + \cos^2 t = 1$$

$$\sin^2 t = 1 - \cos^2 t \qquad \text{Solve for } \sin^2 t$$

$$\sin t = \pm\sqrt{1 - \cos^2 t} \qquad \text{Take square roots}$$

Since $\sin t$ is negative in Quadrant III, the negative sign applies here. Thus

$$\tan t = \frac{\sin t}{\cos t} = \frac{-\sqrt{1 - \cos^2 t}}{\cos t}$$

3. Using Even and Odd properties of Trig Functions

(a) If the secant of angle ***t*** is 2, what is the secant of ***−t*** ?

(b) If sec(x) = 2, what is **−sec(x)**?

(c) If the cotangent of angle t is $\sqrt{3}$, what is the cotangent of ***−t***?

ACTIVITY 3

3. Using Even and Odd properties of Trig Functions

(a) If the secant of angle ***t*** is 2, what is the secant of **−*t*** ?

Solution

Secant is an **even** function. The secant of an angle is the same as the secant of its opposite. So if the secant of angle t is 2, the secant of −t is also 2.

(b) If sec(x) = 2, what is **−sec(x)**?

Secant is an **even** function.

$$sec(-x) = 2$$
$$-sec(x) = -2$$

(c)If the cotangent of angle t is $\sqrt{3}$, what is the cotangent of **−*t***?

Solution

Cotangent is an **odd** function. The cotangent of –t is $-\sqrt{3}$

SOLUTION

4. Finding all Trigonometric functions from the value of one

Find the values of the trigonometric functions of *t* from the given information.

a) $\cos t = -\frac{7}{25}$, terminal point of t is in Quadrant III

b) $\tan t = \frac{1}{4}$, terminal point of t is in Quadrant III

c) $\csc t = 5$, $\cos t < 0$

d) $\tan t = -4$, $\csc t > 0$

ACTIVITY 4

4a. Finding all Trigonometric functions from the value of one

a) $\cos t = -\frac{7}{25}$, terminal point of t is in Quadrant III

$$\sin^2 t + \cos^2 t = 1 \qquad \Rightarrow \sin t = \pm\sqrt{1-\cos^2 t}$$

In quadrant III, tan and cot are positive,
all the other functions are negative (see table of signs, p.410)

So

$$\sin t = -\sqrt{1-(-\tfrac{7}{25})^2} = -\sqrt{\frac{625-49}{625}}$$

$$= -\sqrt{\frac{576}{625}} = -\frac{24}{25}$$

$$\tan t = \frac{\sin t}{\cos t} = \frac{-\frac{24}{25}}{-\frac{7}{25}} = \frac{24}{7}$$

$$\sec t = \frac{1}{\cos t} = -\frac{25}{7}$$

$$\csc t = \frac{1}{\sin t} = -\frac{25}{24}$$

$$\cot t = \frac{1}{\tan t} = \frac{7}{24}$$

SOLUTION

4b. Finding all Trigonometric functions from the value of one

b) $\tan t = \frac{1}{4}$, terminal point of t is in Quadrant III

$$\tan t = \frac{1}{4}$$

$\because$ t terminates in $QIII$ $\therefore$ $\sec t$ is negative

$$\sec^2 t = 1 + \tan^2 t$$
$$\sec t = -\sqrt{1 + \tan^2 t}$$
$$= -\sqrt{1 + \left(\frac{1}{4}\right)^2} = \sec t = -\frac{\sqrt{17}}{4}$$

$$\cos t = \frac{1}{\sec t} = -\frac{4\sqrt{17}}{17}$$

$$\sin t = \cos t \times \tan t = -\frac{\sqrt{17}}{17}$$

$$\csc t = \frac{1}{\sin t} = -\sqrt{17}$$

$$\cot t = \frac{1}{\tan t} = 4$$

SOLUTION

4c. Finding all Trigonometric functions from the value of one

c) $\csc t = 5, \quad \cos t < 0$

$\csc t = 5$ and $\cos t < 0$, so t is in quadrant II. Thus, $\sin t = \frac{1}{5}$ and the terminal point determined by t is $P\left(x, \frac{1}{5}\right)$. Since P is on the unit circle, $x^2 + \left(\frac{1}{5}\right)^2 = 1$. Solving for x gives $x = \pm\sqrt{1 - \frac{1}{25}} = \pm\frac{2\sqrt{6}}{5}$. Since the terminal point is in quadrant II, $x = -\frac{2\sqrt{6}}{5}$ and the terminal point is $P\left(-\frac{2\sqrt{6}}{5}, \frac{1}{5}\right)$. Thus, $\sin t = \frac{1}{5}$, $\cos t = -\frac{2\sqrt{6}}{5}$, $\tan t = -\frac{1}{2\sqrt{6}} = -\frac{\sqrt{6}}{12}$, $\sec t = -\frac{5}{2\sqrt{6}} = -\frac{5\sqrt{6}}{12}$, $\cot t = -2\sqrt{6}$.

SOLUTION

4d. Finding all Trigonometric functions from the value of one

d) $\tan t = -4, \quad \csc t > 0$

$$\tan t = -4$$

$\because \csc t > 0 \;\; \& \tan t < 0$ $\quad\quad \therefore$ $\sec t$ is negative

$$\sec^2 t = 1 + \tan^2 t$$
$$\sec t = -\sqrt{1 + \tan^2 t}$$
$$= -\sqrt{1 + (-4)^2} = \sec t = -\sqrt{17}$$

$$\cos t = \frac{1}{\sec t} = -\frac{\sqrt{17}}{17}$$

$$\sin t = \cos t \times \tan t = \frac{4\sqrt{17}}{17}$$

$$\csc t = \frac{1}{\sin t} = \frac{\sqrt{17}}{4}$$

$$\cot t = \frac{1}{\tan t} = -\frac{1}{4}$$

SOLUTION

5 Writing One Trigonometric Function in Terms of Another

Write the first expression in terms of the second if the terminal point determined by t is in the given quadrant

a) cos t, sin t ; Quadrant IV

b) tan t, cos t ; Quadrant III

c) csc t, cot t ; Quadrant III

d) sin t, sec t ; Quadrant IV

ACTIVITY 5

5a Writing One Trigonometric Function in Terms of Another

Write the first expression in terms of the second if the terminal point determined by t is in the given quadrant

a) cos *t*, sin *t* ; Quadrant IV

$$\sin^2 t + \cos^2 t = 1 \qquad \Rightarrow \cos t = \pm\sqrt{1 - \sin^2 t}$$

Determine the sign:
in quadrant IV, cosine is positive

$$\cos t = +\sqrt{1 - \sin^2 t}$$

SOLUTION

5b Writing One Trigonometric Function in Terms of Another

b) **tan *t*, cos *t* ; Quadrant III**

$\tan t = \dfrac{\sin t}{\cos t}$, and from

$\sin^2 t + \cos^2 t = 1 \qquad \Rightarrow \sin t = \pm\sqrt{1 - \cos^2 t}$

Determine the sign:
in quadrant III, sine is negative

$\sin t = -\sqrt{1 - \cos^2 t}$

So

$$\tan t = \frac{\sin t}{\cos t}$$

$$\tan t = -\frac{\sqrt{1 - \cos^2 t}}{\cos t}$$

(since the denominator, $\cos t$ is negative in q.III,
it will work out for tan to be positive in quadrant III, as the table states)

SOLUTION

5c Writing One Trigonometric Function in Terms of Another

c) csc *t*, cot *t* ; Quadrant III

$$1 + \cot^2 t = \csc^2 t \qquad \Rightarrow \csc t = \pm\sqrt{1 + \cot^2 t}$$

By the sign table, Q.III: $\csc t = -\sqrt{1 + \cot^2 t}$

SOLUTION

5d Writing One Trigonometric Function in Terms of Another

d) sin *t*, sec *t* ; Quadrant IV

$\sin^2 t + \cos^2 t = 1$ $\qquad \Rightarrow \sin t = \pm\sqrt{1 - \cos^2 t}$

$\ldots$reciprocal: $\cos^2 t = \dfrac{1}{\sec^2 t}$ ... so

$$\sin t = \pm\sqrt{1 - \frac{1}{\sec^2 t}}$$

By the sign table, Q.IV: $\quad \sin t = -\sqrt{1 - \dfrac{1}{\sec^2 t}}$

SOLUTION